Reconstruction for Principle C in Russian topicalization: An experimental study[1]

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Introduction. Principle C — stating that a referential expression must be free — belongs to the core of the traditional binding theory; it has been mostly considered to be a universal and purely grammatical constraint. Although in the cases of surface Principle C violation such as (1a), cataphora may serve as a confounding factor, the unacceptability of Principle C violations cannot be fully attributed to such processing limitations: in (1b), the pronoun linearly follows the R-expression, but the bound reading is still unacceptable (grammaticality judgements from Lebeaux 1991). This comes from the fact that the position of the *wh*-phrase that includes the R-expression is derived via Ā-movement, which is commonly argued to reconstruct for binding — in other words, Principle C violation in the configuration prior to *wh*-movement disallows binding in (1b).

(1) a. * He$_i$ likes those pictures of John$_i$.
b. ?* [Which pictures of John$_i$] does he$_i$ like [~~which pictures of John$_i$~~]?

Problem. While reconstruction for Principle C is generally viewed to be universal and is commonly used as an A/Ā-diagnostic (see e.g. van Urk 2015 for discussion), the empirical data are far less clear than expected. Although the transparent contrast coming from the linguists' introspection is hardly reproduced in any of the present experimental studies, the interpretation of the results varies substantially: from the claims that the experimental data are in full accordance with the theoretical predictions (Stockwell et al. 2021), to the complete rejection of Principle C reconstruction (Adger et al. 2017; Bruening & Al Khalaf 2019), with some suggestions that Principle C is a 'soft' violable constraint in the middle (Salzmann et al. 2023; Szarvas 2025). The existing experimental studies mainly focus on English and German — more precisely, on *wh*-movement of DPs and APs containing the target R-expressions, as well as relativization of the former. Seeking to broaden the empirical scope, we report on an experimental study investigating whether topicalization in Russian reconstructs for Principle C.

Data. It is commonly believed that topicalization in Russian OSV clauses is an instance of Ā-movement: the movement of O does not create new configurations for anaphor binding and produces Weak Crossover effects (Bailyn 2004, 2011; Slioussar 2011; Antonyuk 2021; Pereltsvaig 2021, 2022, a.o.). Although reconstruction for Principle C has not been widely used in the previous literature to prove the Ā-status of the O in OSV in Russian, this diagnostic is expected to pattern with the others: topicalization is predicted to reconstruct. We tested this hypothesis experimentally. Our study focused on topicalization of object *that*-clauses; the target R-expression and pronoun occupied the subject positions in the embedded and matrix clauses. The argument status of such *that*-clauses, as well as that of subjects, is uncontroversial, which makes the potential lack of reconstruction inexplicable by the late merger of adjuncts (Lebeaux 1991; see recent discussion in Sportiche 2019). The design included three binary variables: (i) presence of an additional morphologically appropriate antecedent of the target pronoun, (ii) Principle C violation in the base position, (iii) cataphoric dependency between the target pronoun and R-expression. Example stimuli are given in Table 1, conditions 6–8 follow trivially. The first variable was intended to control for processing biases: on the one hand, speakers generally prefer linking the pronoun to some antecedent present in the context (Gordon & Hendrick 1997); this may make them more willing to accept the bound reading. The presence of another appropriate antecedent eliminates this issue but creates another one: in this case, the choice between two antecedents may be conditioned not solely by syntactic considerations, but

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by the pragmatic ones as well. In the previous studies, different decisions concerning this factor were made, which may have influenced the results (see discussion in Szarvas 2025); in our experiment, we opted for including both possibilities as a factor. An additional antecedent (either morphologically appropriate or not) was introduced at the beginning of the sentence by a phrase *po slovam X* / *esli verit' X* 'according to X'. Regarding the two remaining variables, {+C_viol, +cat} conditions featured a surface Principle C violation — the bound reading was expected to fail; in {+C_viol, –cat}, the R-expression was only bound by the pronoun in the base position — if topicalization reconstructs, the coindexing between them has to be banned. The other conditions served as controls: {–C_viol, +cat} were intended to check whether cataphoric dependencies are generally acceptable in Russian — in other words, to what extent the results for {+C_viol, +cat} are explained by Principle C itself rather than by a bias against cataphora. In {–C_viol, –cat} sentences, nothing should have prevented the participants from choosing the bound reading. The experimental task was formulated as a forced-choice comprehension question regarding the reference of the target pronoun; e.g. for condition 2 in Table 1, the question was 'Who remembered something?'. The answer options were as follows: A. additional antecedent (*Artem* for condition 2), B. target R-expression (*Igor*), C. either of them, D. someone else. As the goal of the study was to establish whether the target R-expression and pronoun could be coindexed, the dependent variable was treated as binary: initial options B and C corresponded to the value 'coindexing is possible', and options A and D — to the value 'coindexing is impossible'. 103 native speakers of Russian were included in the sample; each participant was presented with 32 stimuli along with 32 fillers which had an unambiguously correct answer. The data were analyzed in R (R Core Team 2020) using generalized linear mixed-effects model with a *binomial* family (lme4 package). The answer distributions for initial and binary variables are given in Fig. 1 and 2 respectively.

Results and discussion. According to our results, Principle C expectedly blocks binding in the 'surface' conditions 1 and 5. This cannot be reduced to a bias against cataphora: although the cataphoric antecedent is indeed chosen less often than the anaphoric one in conditions without Principle C violation (cf. 3 and 4), it is still possible and even preferred. Regarding the conditions with topicalization, in the absence of an alternative antecedent, the pronoun is almost always coindexed with the target R-expression, in contrast to the surface Principle C condition (cf. 6 and 5). This result cannot solely be explained by a processing bias towards the antecedent present in the context: coindexing with the target R-expression is preferred even when an additional appropriate antecedent is given (cf. 2). Although binding is still less accessible in the reconstruction conditions than in controls (cf. 2 and 4, 6 and 8), this is not enough to argue for obligatory reconstruction coupled with a 'hard' Principle C constraint. In our talk, we will compare our data with the results of the previous experimental studies, as well as present two additional experiments, and discuss the potential implications for the theory.

1	+ant	+C_viol	+cat	*Po slovam Artema, on vspomnil, čto Igor' gotovit otčet dlja soveščanija.* According to Artem(M), he remembered that Igor(M) was preparing a report for the meeting.
2	+ant	+C_viol	–cat	*Po slovam Artema, čto Igor' gotovit otčet dlja soveščanija, on vspomnil.* According to Artem(M), that Igor(M) was preparing a report for the meeting, he remembered.
3	+ant	–C_viol	+cat	*Po slovam Artema, čto on gotovit otčet dlja soveščanija, Igor' vspomnil.* According to Artem(M), that Igor(M) was preparing a report for the meeting, he remembered.
4	+ant	–C_viol	–cat	*Po slovam Artema, Igor' vspomnil, čto on gotovit otčet dlja soveščanija.* According to Artem(M), Igor(M) remembered that he was preparing a report for the meeting.
5	–ant	+C_viol	+cat	*Po slovam Niny, on vspomnil, čto Igor' gotovit otčet dlja soveščanija.* According to Nina(F), he remembered that Igor(M) was preparing a report for the meeting.
				…

Table 1. Example stimuli

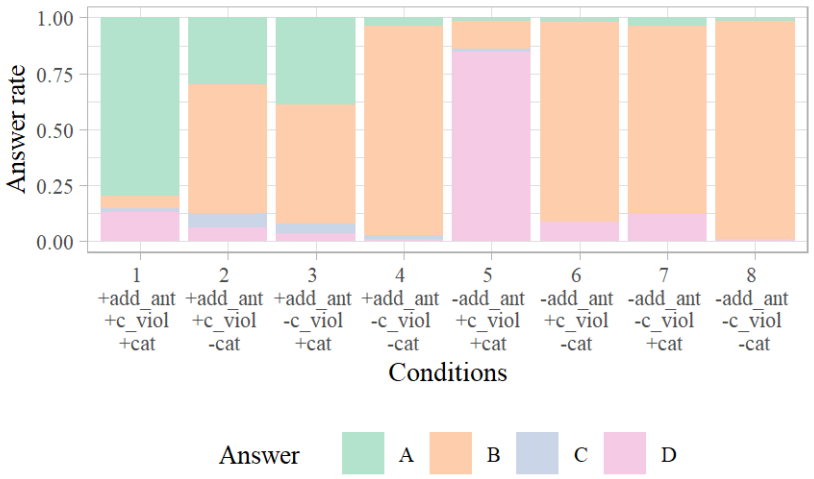


Fig. 1. Results (initial)

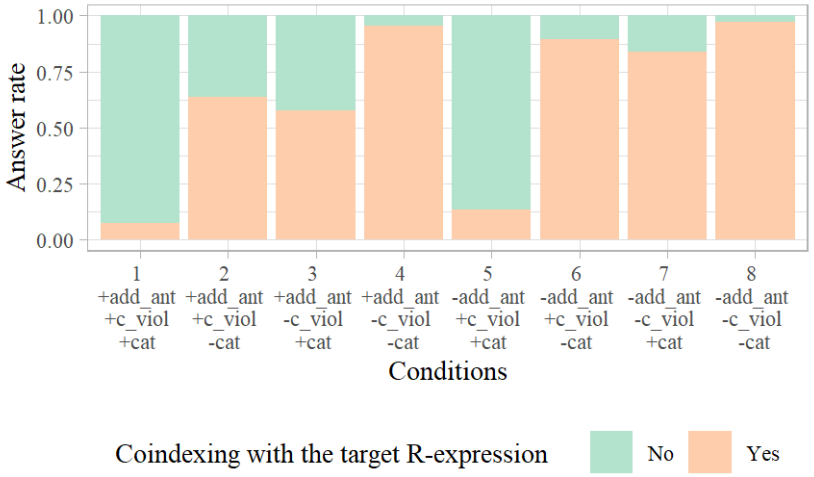


Fig. 2. Results (binary variable)

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